

DENNIS MWANGI

Data Scientist & Economist • Machine Learning, Statistics & Dashboard Development

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PROFESSIONAL SUMMARY

Economics & Statistics graduate (Chuka University, expected Nov 2026) with hands-on experience in data cleaning, statistical modeling, machine learning, and dashboard development. Proficient in Python, R, SQL, Power BI, and Tableau, with a track record of turning large, messy datasets into predictive models and decision-ready dashboards across healthcare, financial, and marketing analytics. Combines an economist's grounding in causal reasoning and econometrics with a data scientist's ability to build, validate, and deploy machine learning models end to end.

TECHNICAL SKILLS

Programming & Query Languages: Python, R, SQL

Statistical & Analytical Methods: Machine Learning, Statistical Modeling, Regression, Econometrics, Bayesian Statistics, Time Series Forecasting, Marketing Mix Modeling & Optimization

Data Management: Data Cleaning & Preparation, Databases, Big Data, Git/GitHub

Visualization & BI Tools: Power BI, Tableau, Streamlit, R Shiny (Posit), Jupyter Notebook, Excel

Other: Generative AI tools, Microsoft Office

EXPERIENCE

Freelance Data Analyst

2025 — Present

Independent

- Cleaned, prepared, and merged large datasets from multiple sources using Python, SQL, and Excel to ensure analysis-ready data quality.
- Built regression and econometric models to assess marketing metrics and inform client strategy, including a marketing mix model quantifying the impact of TV, digital, and promotional spend on auto sales.
- Implemented Marketing Mix Modeling and optimization frameworks to evaluate campaign effectiveness and guide budget allocation.
- Applied Bayesian statistics to improve the predictive accuracy of forecasting models.
- Designed interactive dashboards in Power BI and Tableau to present insights and support data-driven decision-making.

Team Lead, Data Visualization Internship

July 1 – July 31, 2026

Excellerate Internship Program

- Led a team of interns through the design and delivery of an interactive analytics dashboard, from data cleaning through to final visualization.
- Directed task allocation, code review, and quality checks to keep the project on schedule and consistent in design.
- Coordinated the build of the Team 13 Opportunity Dashboard, a Streamlit application translating raw opportunity data into a filterable, decision-ready tool.
- Earned a Data Visualization certificate on completion of the program.

SELECTED PROJECTS

Hypertension Risk Prediction System *(Python, Streamlit, R Shiny)*

- Cleaned and preprocessed a clinical dataset, handling missing values and encoding categorical variables; built and evaluated logistic regression and random forest models with scikit-learn, tuning hyperparameters via cross-validation.
- Deployed as both a Streamlit app and a professional R Shiny (Posit) application with model comparison and patient-specific predictions.

Customer Churn & Fraud Detection Models *(Python, Streamlit)*

- Built classification models to flag at-risk customers for churn and to distinguish fraudulent from legitimate transactions, each deployed as a live, interactive Streamlit application.

Olympic Medal Predictor *(Python, Streamlit)*

- Applied EDA, feature engineering, and machine learning to estimate Olympic medal outcomes from historical performance data.

Marketing Mix Model — Auto Sales Analysis *(R, SQL, Excel)*

- Aggregated sales, pricing, and marketing-spend data and built a regression/time-series model quantifying channel-level impact on sales.

EDUCATION & CERTIFICATIONS

B.Sc. Economics and Statistics

Chuka University

Second Class Upper Division · Expected graduation: November 21, 2026

- Professional Certificate, Data Analytics/Data Science — CDAM (Center of Data Analytics and Modeling)
- Generative AI and Microsoft Power BI Certificate — Exodus Experts
- Data Visualization Certificate — Excellerate Internship Program

Languages: English (Fluent) · Swahili (Fluent)